REAL ANALYSIS II (MAT5CJ301) · SEMESTER 5 · MODULE 2 · MODEL PAPER

# Differentiation

Bartle & Sherbert §6.1 (6.1.1–6.1.7), §6.2 (6.2.1–6.2.13) · FYUGP pattern (April 2026), doubled

**HOW THIS PAPER WORKS**

A real FYUGP paper has 10 short (3 marks), 8 paragraph (6 marks) and 2 essay (10 marks, answer one) questions. This paper has **twice as many** (20 + 16 + 4), all from Module 2. Each of the module's 24 concepts is the main point of exactly one question; 16 of the finer points get a question of their own.

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| **Time** | 4 hours (or two sittings of 2 hours) |
| **Maximum** | 140 marks (A ceiling 48, B ceiling 72, C answer any two) |

No old Calicut paper has asked a Module 2 question (differentiation used to be taught in an earlier semester). So the tags are: **CALC-24 Q5** = related FYUGP Calculus paper (MAT1MN102, Nov 2024), question 5; **PYQ MTS-22 MCQ 5** = MTS 6B 10, March 2022, MCQ 5, "adapted" = a 1-mark MCQ turned into a 3-mark question; **Bartle Ex 6.1.1(b)** = a starred exercise of the textbook; **Bartle Thm 6.2.4** = a theorem of the textbook. Unless a question says otherwise, you may quote the Plus Two derivatives ($\sin' = \cos$, $\cos' = -\sin$, $(e^x)' = e^x$, $(\ln x)' = 1/x$).

## Section A · Short answer

All questions can be answered. Each carries 3 marks. Ceiling 48.

**1.** Define the derivative of $f : I \to \mathbb{R}$ at $c \in I$ using $\varepsilon$ and $\delta$. Use your definition to verify that the derivative of $f(x) = x^2$ at $c = 2$ is 4.

**Bartle Def 6.1.1**

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**2.** Using the definition of the derivative, find $f'(c)$ for $f(x) = 3x$, $x \in \mathbb{R}$.

**CALC-24 Q5 · related FYUGP**

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**3.** From the definition, find the derivative of $g(x) = 1/x$ at a point $c \neq 0$.

**Bartle Ex 6.1.1(b)**

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**4.** If $f$ is differentiable at $c$, show that $f'(c) = \lim_{n\to\infty} n\{f(c + 1/n) - f(c)\}$. Show by $f(x) = |x|$ that the converse is false.

**Bartle Ex 6.1.13**

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**5.** Show that $f(x) = x^{1/3}$ is continuous but not differentiable at 0. What does the graph do at 0?

**Bartle Ex 6.1.2**

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**6.** State the quotient rule and use it to differentiate $f(x) = \dfrac{x}{1 + x^2}$.

**Bartle Ex 6.1.5(a)**

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**7.** Prove by induction, using the product rule, that $D\,x^n = nx^{n-1}$ for $n \in \mathbb{N}$. Deduce $D\,x^{-n} = -nx^{-n-1}$.

**Bartle Cor 6.1.4**

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**8.** State the chain rule. Differentiate $g(x) = \sqrt{x^2 - 2x + 5}$ and say why it is differentiable for every $x \in \mathbb{R}$.

**Bartle Ex 6.1.5(b)**

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**9.** Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable. Prove that $f'$ is odd if $f$ is even, and $f'$ is even if $f$ is odd.

**Bartle Ex 6.1.9**

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**10.** If $f$ is differentiable at $c$ and $f(c) \neq 0$, find $|f|'(c)$. If $f(c) = 0$, when is $|f|$ differentiable at $c$? Find $|f|'$ for $f(x) = x^2 - 1$ and the points where it fails.

**Bartle Ex 6.1.7**

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**11.** Show that $h(x) = x|x|$ is differentiable at every $x \in \mathbb{R}$ and find $h'$.

**Bartle Ex 6.1.8(c)**

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**12.** Using the quotient rule, $\sin' = \cos$ and $\cos' = -\sin$, prove $D \tan x = \sec^2 x$ and $D \sec x = \sec x \tan x$. State where they hold.

**Bartle 6.1.7(d)**

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**13.** For $r > 0$ let $f(x) = x^r \sin(1/x)$ for $x \neq 0$, $f(0) = 0$. For which $r$ does $f'(0)$ exist?

**Bartle Ex 6.1.12**

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**14.** Find all critical points of $f(x) = x^3 - 3x + 1$. Which of them can be relative extrema, and why? State the theorem you use.

**CALC-24 Q10 · related FYUGP**

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**15.** Find the relative extrema of $f(x) = \dfrac{x^2}{x^4 + 16}$ using the First Derivative Test.

**CALC-24 Q18 · related FYUGP**

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**16.** Verify the Mean Value Theorem for $f(x) = x^3$ on $[0, 2]$ and find the number $c$.

**Bartle Thm 6.2.4**

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**17.** Define "increasing" and "strictly increasing" for $f : I \to \mathbb{R}$ (as in Bartle). Give an example of a strictly increasing function whose derivative is 0 at a point.

**CALC-24 Q9 · related FYUGP**

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**18.** Find the intervals on which $f(x) = x^4 - 5x^3 + 9x^2$ is increasing and decreasing.

**CALC-24 Q20(1) · related FYUGP**

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**19.** Using the sign of the derivative, decide which of the following are increasing on $(0, 1)$: (i) $1 - x^3$, (ii) $x^2 + x$, (iii) $1/x^2$.

**PYQ MTS-22 MCQ 5 · adapted**

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**20.** State Lemma 6.2.11 (the sign of $f(x) - f(c)$ near $c$ when $f'(c) \neq 0$) and prove part (a).

**Bartle Lemma 6.2.11**

## Section B · Paragraph / problem

All questions can be answered. Each carries 6 marks. Ceiling 72.

**21.** Prove that a function differentiable at $c$ is continuous at $c$. Show by an example that the converse is false.

**Bartle Thm 6.1.2**

**related MAT-22 MCQ 4, MAT-21 MCQ 3 ($\lim |x|/x$)**

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**22.** Let $f(x) = x^2$ for $x \le 1$ and $f(x) = 4\sqrt{x}$ for $x > 1$. (a) Is $f$ differentiable at 1? Justify. (b) Find constants $a, b$ such that $F(x) = x^2$ ($x \le 1$), $F(x) = ax + b$ ($x > 1$) is differentiable at 1. (c) Is $G(x) = x^2$ ($x \le 1$), $G(x) = \sqrt{x}$ ($x > 1$) differentiable at 1?

**CALC-24 Q13 · related FYUGP**

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**23.** State and prove the product rule for derivatives at a point. Deduce $(1/g)'(c) = -g'(c)/g(c)^2$ when $g(c) \neq 0$.

**Bartle Thm 6.1.3(c)**

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**24.** State and prove Carathéodory's Theorem. Find $\varphi$ for $f(x) = x^3$ at $c$ and deduce $f'(c)$. Explain why $|x|$ has no such $\varphi$ at 0.

**Bartle Thm 6.1.5**

**25.** Let $f(x) = x^2 \sin(1/x)$ for $x \neq 0$, $f(0) = 0$. Show that $f$ is differentiable at every point, find $f'$, and show that $f'$ is not continuous at 0.

**Bartle Ex 6.1.7(e)**

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**26.** State and prove the Interior Extremum Theorem. Show by examples that "interior", "differentiable" and the converse cannot be dropped.

**Bartle Thm 6.2.1**

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**27.** State and prove Rolle's Theorem. Verify it for $f(x) = x(x-1)(x-2)$ on $[0, 2]$.

**Bartle Thm 6.2.3**

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**28.** Prove that a function continuous on $[a, b]$ with $f' = 0$ on $(a, b)$ is constant. Deduce that two such functions with equal derivatives differ by a constant. Why does the argument fail on $(0, 1) \cup (2, 3)$?

**Bartle Thm 6.2.5, Cor 6.2.6**

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**29.** State and prove part (a) of the First Derivative Test. Apply it to $f(x) = x^3 - 3x$, and to $f(x) = |x|$ at 0.

**Bartle Thm 6.2.8**

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**30.** (a) Use Rolle's Theorem to show that between two consecutive zeros of $\sin x$ there is a zero of $\cos x$. (b) Show that $p(x) = x^3 + 3x + 1$ has exactly one real root and locate it in an interval of length $\frac{1}{4}$.

**Bartle 6.2.9(a)** **related MAT-21 Q17**

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**31.** Use the Mean Value Theorem for $f(x) = \sqrt{x}$ on $[100, 105]$ to show $10.2272 < \sqrt{105} < 10.25$, and then sharpen the lower bound.

**Bartle 6.2.9(b)**

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**32.** Using the Mean Value Theorem prove (a) $e^x \geq 1 + x$ for all $x \in \mathbb{R}$, with equality only at $x = 0$; (b) $\dfrac{x-1}{x} < \ln x < x - 1$ for $x > 1$.

**Bartle 6.2.10(a), Ex 6.2.7**

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**33.** Prove (a) $-x \leq \sin x \leq x$ for $x \geq 0$; (b) $|\sin x - \sin y| \leq |x - y|$ for all $x, y$; (c) if $f$ is differentiable on an interval $I$ and $|f'| \leq K$ on $I$, then $f$ is Lipschitz.

**Bartle 6.2.10(b), Ex 6.2.15** **related MTS-22 MCQ 4**

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**34.** If $\alpha > 1$, prove Bernoulli's inequality $(1 + x)^\alpha \geq 1 + \alpha x$ for all $x > -1$, with equality if and only if $x = 0$.

**Bartle 6.2.10(c)**

**35.** Let $0 < \alpha < 1$. Show $x^\alpha \le \alpha x + (1-\alpha)$ for $x \ge 0$, and deduce $a^\alpha b^{1-\alpha} \le \alpha a + (1-\alpha)b$ for $a, b > 0$. Hence prove $\sqrt{ab} \le \dfrac{a+b}{2}$.

**Bartle 6.2.10(d)**

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**36.** Show that $g : [-1,1] \to \mathbb{R}$ with $g = -1$ on $[-1,0)$, $g(0) = 0$, $g = 1$ on $(0,1]$ is not the derivative of any function. Give two functions on $\mathbb{R}$, not differing by a constant, whose derivative is $h(x) = 0$ ($x < 0$), $h(x) = 1$ ($x > 0$) for all $x \neq 0$.

**Bartle Cor 6.2.13, Ex 6.2.12**

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## Section C · Essay

Answer any two. Each carries 10 marks.

**37.** State and prove the Chain Rule using Carathéodory's Theorem. Explain why the naive proof (dividing by $f(x) - f(c)$) fails. Use the rule to differentiate $\sin(x^2)$ and $|x^2 - 1|$.

**Bartle Thm 6.1.6**

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**38.** State and prove the Mean Value Theorem. Give its geometric meaning and show, with examples, that neither continuity on $[a, b]$ nor differentiability on $(a, b)$ can be dropped.

**Bartle Thm 6.2.4**

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**39.** State and prove Darboux's Theorem (you may use Lemma 6.2.11). Treat both orderings $f'(a) < k < f'(b)$ and $f'(a) > k > f'(b)$.

**Bartle Thm 6.2.12**

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**40.** Let $f$ be differentiable on an interval $I$. Prove: $f$ is increasing if and only if $f' \geq 0$ on $I$. Show that $f' > 0$ implies strictly increasing but not conversely, and that $f'(c) > 0$ at a single point does not give monotonicity near $c$. Apply the result to $f(x) = x - \sin x$.

**Bartle Thm 6.2.7, Ex 6.2.10, 6.2.13**

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# Coverage

Topic codes match the prepare file (T01–T24, one per concept K01–K24 of the brief).

| TOPIC | QUESTIONS |
|---|---|
| T01 Derivative at a point | 1 (main), 2, 3, 4 |
| T02 Differentiable ⇒ continuous | 21 (main), 5, 22 |
| T03 Algebra of derivatives | 23 (main), 6 |
| T04 Power rule, sums, products | 7 |
| T05 Carathéodory | 24 |
| T06 Chain rule | 37 (main), 8, 9 |
| T07 Derivative of $\lvert f\rvert$ | 10 (main), 11 |
| T08 tan, sec, cot, csc | 12 |
| T09 $x^2\sin(1/x)$ | 25 (main), 13 |
| T10 Interior Extremum Theorem | 26 (main), 14 |
| T11 Rolle | 27 |
| T12 Mean Value Theorem | 38 (main), 16 |
| T13 $f' = 0$ ⇒ constant | 28 |
| T14 Monotonicity | 40 (main), 17, 18, 19 |
| T15 First Derivative Test | 29 (main), 15 |
| T16 Roots by Rolle | 30 |
| T17 Approximation by MVT | 31 |
| T18 $e^x$ and $\ln x$ inequalities | 32 |
| T19 $\sin x$, Lipschitz | 33 |
| T20 Bernoulli | 34 |
| T21 Weighted AM–GM | 35 |
| T22 Local sign lemma | 20 |

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|---|---|
| T23 Darboux | 39 |
| T24 Not a derivative | 36 |

**All 7 past/related questions appear**: CALC-24 Q5 (Q2), Q9 (Q17), Q10 (Q14), Q13 (Q22), Q18 (Q15), Q20(1) (Q18), and PYQ MTS-22 MCQ 5 (Q19). The old CBCSS papers have no Module 2 question; the other questions come from Bartle's theorems and starred exercises.

**Printed question check.** In CALC-24 Q13 the second branch is $4\sqrt{x}$, so $f(1) = 1$ but $\lim_{x\to 1^+} f = 4$: $f$ is not continuous at 1, hence not differentiable there.

# Question number → topic

| Q | TOPIC | Q | TOPIC |
|---|---|---|---|
| 1 | T01 | 21 | T02 |
| 2 | T01 | 22 | T02 |
| 3 | T01 | 23 | T03 |
| 4 | T01 | 24 | T05 |
| 5 | T02 | 25 | T09 |
| 6 | T03 | 26 | T10 |
| 7 | T04 | 27 | T11 |
| 8 | T06 | 28 | T13 |
| 9 | T06 | 29 | T15 |
| 10 | T07 | 30 | T16 |
| 11 | T07 | 31 | T17 |
| 12 | T08 | 32 | T18 |
| 13 | T09 | 33 | T19 |
| 14 | T10 | 34 | T20 |
| 15 | T15 | 35 | T21 |
| 16 | T12 | 36 | T24 |

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|---|---|---|---|
| 17 | T14 | 37 | T06 |
| 18 | T14 | 38 | T12 |
| 19 | T14 | 39 | T23 |
| 20 | T22 | 40 | T14 |